

Energy between 3 to 2 = 1.9 eV. Hence, answer is C.

29 Ans: B

$$KE = \frac{1}{2} mv^2$$

$$m(KE) = \frac{1}{2} m^2 v^2 = \frac{1}{2} p^2$$

$$p = \sqrt{2m(KE)}$$

$$\text{From de Broglie's equation, } \lambda = h / p = h / \sqrt{2m(KE)}$$

Hence, we can see that as KE increases, λ decreases, and it will not be linear.

30 Ans: D